

Revisions to Wainwright’s Chapter 15 Minimax Lower Bounds in the `StatLean.Minimaxity` Formalization

Formalization notes — StatLean

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Abstract

We formalized the minimax lower bounds of Wainwright, *High-Dimensional Statistics: A Non-Asymptotic Viewpoint* (Cambridge, 2019), Chapter 15. Several results were stated or proved differently from the book. This note explains *why*, sorting each revision into one of four causes: (i) genuine *informal imprecision* in the book (loss convention; “over the class” richness); (ii) *finite-sample edge guards* the book legitimately omits from an asymptotic statement; (iii) *looser constants* forced by infrastructure missing from Mathlib; and (iv) *proof-strategy substitutions* that avoid machinery absent from Mathlib while proving the same statement. The short answer to “is Wainwright sloppy, or did you revise for tractability?” is: *mostly the latter*. The book’s mathematics is correct; only two revisions trace to genuine informal looseness (the loss convention of Examples 15.8/15.11, and the implicit richness of “the class”). One result (Example 15.11, the optimal $n^{-4/9}$ quadratic-functional rate) was *removed* rather than shipped with a vacuous proof.

Contents

1 Overview and the central dichotomy

The full area `StatLean.Minimaxity` builds with 0 sorry. Table ?? lists every revision against the verbatim book statement.

Book result	Revision	Cause	§
Ex 15.5 (uniform loc.)	add $n \geq 2$	(ii) finite- n edge	??
Ex 15.7 (Lipschitz dens.)	existential witness; domain $[-\frac{1}{2}, \frac{1}{2}]$	(i)/(stub)	??
Ex 15.8 (quad. two-pt)	absolute loss $\Phi = \text{id}$; real $\int (f')^2$; existential	(i) imprecision	??
Ex 5.12 (Sobolev entropy)	add $\delta \leq \frac{1}{2}$	(ii) (book says “small δ ”)	??
Ex 15.14 (lin. regr.)	add $r \geq 40$; constant $128^{-1} \rightarrow 2560^{-1}$	(ii)+(iii)	??,??
Lemma 15.17 (Gauss. max-ent.)	matched-Gaussian proof; add $\Sigma^j \succ 0$	(iv) tractability	??
Ex 15.19 (PCA)	pairwise-KL proof; $d \geq 40$; $128^{-1} \rightarrow 2048^{-1}$	(iv)+(ii)+(iii)	??,??,??
Ex 15.11 (quad. optimal)	removed	(iv) infeasible	??

Table 1: Revisions and their causes.

2 Genuine informal imprecision in the book

2.1 The loss convention of Examples 15.8 and 15.11

This is the one place where the book is genuinely informally loose. Across Chapter 15 Wainwright silently alternates between *squared* error and *absolute* error. Example 15.4 (Gaussian location) and Example 15.7 (Lipschitz density) are stated for squared error, $\inf_{\hat{\theta}} \sup_{\theta} \mathbb{E}_{\theta}[(\hat{\theta} - \theta)^2] \gtrsim n^{-2/3}$, whereas Example 15.8 (quadratic functional) is stated for *absolute* error:

$$\text{“} \sup_{f \in \mathcal{F}_2} \mathbb{E} |\hat{\theta}(f) - \theta(f)| \gtrsim n^{-1/2} \text{”} \quad (\text{Wainwright, Eq. (15.24)})$$

and likewise Example 15.11, “ $\sup \mathbb{E} |\hat{\theta} - \theta| \geq \delta/4 \asymp n^{-4/9}$ ”. The exponent and the loss *must agree*: for the genuine functional $\theta(f) = \int_0^1 (f')^2$, the squared-error rate is n^{-1} (Ex 15.8) or $n^{-8/9}$ (Ex 15.11), not $n^{-1/2}/n^{-4/9}$.

The original Lean stub used the squared distortion $\Phi = (\cdot)^2$ *together with* the absolute-error exponent $n^{-1/2}$ and a placeholder θ_{func} *decoupled* from the real functional—this is internally inconsistent (it is only satisfiable because the existential constant “ $\exists c$ ” is quantified after n ; see §??). We repaired Example 15.8 to the book-faithful absolute-loss form with the *genuine* functional:

```
theorem quadratic_functional_two_point_rate (n : Nat) (hn : 1 < n) :
  ( : Type) ( _ : MeasurableSpace ) (f : → →) (func : →)
  (Pn : Kernel (Fin n →)) ( _ : IsMarkovKernel Pn),
  ( i, x Set.Icc (0:) 1, (1/2:) f i x)
  ( i, func i = x in Set.Icc (0:) 1, (deriv (f i) x)^2) -- the REAL (f')^2
  ( i, Pn i = Measure.pi fun _ => (volume.restrict (Set.Icc 0 1)).withDensity
    (fun x => ENNReal.ofReal (f i x)))
  c : , 0 < c
  ENNReal.ofReal (c * (n :) ^ (-(1:)/2)) minimaxRiskDist id func Pn -- Φ= id
```

Cause: (i) book imprecision. The mathematics in the book is correct; the looseness is purely notational (which loss is in force), and it is invisible until one writes the loss function explicitly in Lean. *Caveat:* Example 15.14 (linear regression) really is squared error (“ $\mathbb{E}[\frac{1}{n} \|X(\hat{\theta} - \theta)\|_2^2]$ ”, Eq. (15.36)), so there $(\cdot)^2$ is the faithful choice—we kept it.

2.2 “The minimax risk over the class” requires a rich family

Wainwright writes “ $\inf_{\hat{\theta}} \sup_{f \in \mathcal{F}} \dots \gtrsim n^{-2/3}$ ”, meaning the supremum over the *full* class $\mathcal{F}$ of 1-Lipschitz densities. The book *constructs* a specific two-element sub-experiment $\{f_0, g\} \subset \mathcal{F}$ (the hat perturbation, Eq. (15.19)) and lower-bounds the risk by the two-point bound on that pair. A naive Lean stub quantified the bound over an *arbitrary* family $\mathbf{f} : \rightarrow \rightarrow$: that statement is *false*, because a trivial (one-point) ι has Hellinger modulus 0 and hence minimax risk 0. We restructured to the *existential* form, which asserts exactly the witnessing sub-experiment the book builds (a valid lower bound on the full-class minimax), matching how `pca_minimax_rate` and `linear_regression_minimax_rate` are stated:

```
theorem lipschitz_density_pointwise_rate (n : Nat) (hn : 1 < n) :
  ( : Type) ( _ : MeasurableSpace ) ( _ : Nonempty ) (f : → →) ...,
  ( i, LipschitzWith 1 (f i) x, (1/2:) f i x) ...
  c : , 0 < c ENNReal.ofReal (c * (n :) ^ (-(2:)/3)) minimaxRiskDist (·^2) func Pn
```

Cause: (i) formalization precision. The book’s argument is correct; “over the class” silently assumes the class is rich enough to contain the perturbation. The existential form makes that explicit.

A second, smaller fix in the same files: the densities must live on the book’s *bounded* interval ($[-\frac{1}{2}, \frac{1}{2}]$ for Ex 15.7, $[0, 1]$ for Ex 15.8). The stub used the unrestricted Lebesgue **volume**; but $f \geq \frac{1}{2}$ on all of $\mathbb{R}$ has infinite mass, so `IsProbabilityMeasure` is unsatisfiable. This is a *stub* bug, not a book issue—Wainwright is explicit about the domain (“densities on $[-\frac{1}{2}, \frac{1}{2}]$ ”). We use `volume.restrict (Set.Icc ...)`.

3 Finite-sample edge guards (the book is asymptotic)

These revisions add a hypothesis the book reasonably omits because its statements are asymptotic. They are *not* corrections of the book; they pin down the regime in which the displayed bound is true.

3.1 Example 15.5: $n \geq 2$

The book takes a pair with $|\theta - \theta'| = 2\delta := \frac{1}{4n}$ and concludes $\|U_\theta^n - U_{\theta'}^n\|_{TV}^2 \leq H^2 \leq \frac{1}{2}$, hence $\mathbb{E}[(\hat{\theta} - \theta)^2] \geq \frac{1-1/\sqrt{2}}{128} n^{-2}$. At $n = 1$ with the (formalized) separation $\theta_1 = 1/n = 1$, the shifted unit-uniforms `Unif[0, 1]` and `Unif[1, 2]` are *a.e. disjoint*, so $\|\cdot\|_{TV} = 1$ and the two-point factor $1 - TV$ vanishes—the bound is vacuous. We add $n \geq 2$:

```
-- LEAN-ONLY: n 2 (at n=1 the shifted unit-uniforms [0,1],[1,2] are a.e. disjoint TV=1)
theorem uniform_location_minimax_rate (n : Nat) (hn : 2 ≤ n) ...:
  ENNReal.ofReal ((1 - 1 / Real.sqrt 2) / 128 / (n : ℝ)^2) minimaxRiskDist (·^2) id P
```

The constant $\frac{1-1/\sqrt{2}}{128}$ is exactly the book’s; the proof actually delivers the stronger $1 - TV \geq \frac{1}{32}$ via the exact n -fold Hellinger affinity $H_n^2 = 2(1 - \rho_1^n)$ and $\rho_1^n = (1 - 1/n)^n \geq \frac{1}{4}$ for $n \geq 2$. **Cause: (ii).**

3.2 Example 5.12: $\delta \leq \frac{1}{2}$

Here the book itself flags the regime: “ $\log N(\delta; \mathcal{E}, \|\cdot\|_2) \asymp (1/\delta)^{1/\alpha}$ for all suitably small $\delta > 0$.” The Sobolev ellipsoid has ℓ^2 -diameter ≤ 2 , so no δ -separated pair exists once $\delta > 2$. We make “suitably small” concrete as $\delta \leq \frac{1}{2}$ (which our sphere-packing embedding needs), tagged **LEAN-ONLY**. **Cause: (ii); the book is already aware.**

3.3 Examples 15.14 and 15.19: dimension thresholds $r \geq 40$, $d \geq 40$

Fano’s method requires a packing of cardinality $M = e^{\Omega(\dim)}$ for the bracket $1 - \frac{I + \log 2}{\log M}$ to be bounded below; below a constant dimension $\log M$ is too small and the bracket cannot beat a positive right-hand side. The book’s PCA/regression rates are large-dimension statements, so it never states the threshold. We add it (and document that the small- d regime is a separate two-point argument):

```
-- LEAN-ONLY: d d -Fano's method needs eΩ~{d} hypotheses; the small-d regime is a separate bound
theorem pca_minimax_rate {n d : Nat} (hn : 1 ≤ n) (hd0 : 40 ≤ d) ( : ) (h : 0 < ) ...:
  ENNReal.ofReal (2048-1 * min ((1 + ) / ^2 * (d / n)) 1) minimaxRiskDist (·^2) (Subtype.val) P
```

Cause: (ii). (The loosened constant 2048^{-1} is §??.)

3.4 Lemma 15.17: Σ^j positive-definite

The lemma is about *covariance* matrices; positivity is implicit in the book (“conditioned on $J = j$ has a Gaussian distribution with covariance Σ^j ”). The closed-form KL we use requires $\Sigma^j \succ 0$ (else the Gaussian degenerates), so we state (hS : j, (S j).PosDef) as a tagged USER-INPUT. **Cause:** (ii) made explicit.

4 Looser constants forced by missing infrastructure

For Examples 15.14 and 15.19 the book’s leading constant (128^{-1}) is correct but *unreachable from the packing brick available in Mathlib/StatLean*. The sharp sphere-packing bound $\log M \geq n \log 2$ (Example 5.8) requires a spherical-cap surface-measure / Wallis-integral argument absent from Mathlib; the available constructive brick `exists_sphere_packing` certifies only $\log M \geq n/10$. Propagating $r/10$ (resp. $(d-1)/10$) through the local-packing condition (15.35b) forces a smaller separation, hence a looser leading constant: $128^{-1} \rightarrow 2560^{-1}$ for regression and $128^{-1} \rightarrow 2048^{-1}$ for PCA. Per the project convention (CLAUDE.md §1) we state the *provable* constant and document the deviation in-docstring:

```
-- USER-INPUT: constant loosened from book 128-1 to the value provable from the n/10 sphere-packing
brick.
ENNReal.ofReal ((v :) * r / (2560 * n)) minimaxRiskDist (·^2) g P -- book: v·rank/(128 n)
```

Cause: (iii) formalization difficulty, not a book error. The book is right; we prove a weaker constant because the sharp packing constant is itself hard to formalize.

5 Proof-strategy substitutions (same statement, more formalizable)

These do not change the theorem (beyond the items above); they replace the book’s *proof* with one that avoids machinery Mathlib lacks. This is the clearest “revised for tractability” bucket.

5.1 Lemma 15.17 via the matched Gaussian (no differential entropy)

Wainwright proves the log-determinant mutual-information bound from the *differential-entropy* decomposition $I(Z; J) = h(Z) - \frac{1}{M} \sum_j h(Z \mid J=j)$ together with the Gaussian maximum-entropy theorem $h(Z) \leq \frac{1}{2} \log((2\pi e)^d \det \text{cov}(Z))$ (“a consequence of the maximum entropy property of the multivariate Gaussian; see Exercise 15.14”). Mathlib has *no* differential entropy of measures, so this route would require building that theory from scratch.

We avoid it entirely. Since $I(Z; J) = \frac{1}{M} \sum_j D(\mathcal{N}(0, \Sigma^j) \parallel \bar{Q})$ and the mixture $\bar{Q}$ *minimizes the average KL* (the already-closed Exercise 15.11, `sum_klDiv_mixture_le`), comparing instead to the *matched Gaussian* $G = \mathcal{N}(0, \text{cov}(Z))$ gives $I(Z; J) \leq \frac{1}{M} \sum_j D(\mathcal{N}(0, \Sigma^j) \parallel \mathcal{N}(0, \text{cov}(Z)))$, and each term is the closed-form multivariate Gaussian KL; the $(2\pi e)^d$ never appears and a trace identity $\frac{1}{M} \sum_j \text{tr}(\text{cov}(Z)^{-1} \Sigma^j) = d$ cancels the linear terms:

```
-- I(Z;J) = M-1 Σ klDiv (Q) (mixture) M-1 Σ klDiv (Q) G (mixture minimizes avg KL)
exact sum_klDiv_mixture_le (fun j => Q j) G -- G = (0, covZ)
-- each term, closed form (no (2e)^d):
rw [hQ j, hG_def, klDiv_multivariateGaussian_zero (S j) covZ (hS j) hcovPD]
```

The only genuinely new **ForMathlib** content is the multivariate zero-mean Gaussian KL $D(\mathcal{N}(0, A) \parallel \mathcal{N}(0, B)) = \frac{1}{2}(\log \det B - \log \det A + \text{tr}(B^{-1}A) - d)$ (**GaussianKLMulti.lean**: KL map-invariance + linear push-forward + spectral diagonalization + the 1-D unequal-variance KL). **Cause:** (iv).

5.2 Example 15.19 (PCA) via the pairwise bound (no sphere isotropy)

The book bounds the mutual information with Lemma 15.17 and then *averages over a random orthonormal matrix* U to exploit the isotropy of the uniform measure on the sphere, $\mathbb{E}_U[(U\Delta_j)(U\Delta_j)^\top] = \frac{1}{d-1}I$ (the step producing “ $\Gamma_{\text{low}} = (1 + \frac{\delta^2\nu}{d-1})I_{d-1}$ ”). Formalizing Haar measure on $O(d-1)$ / uniform-sphere second moments is heavy.

We replace this with the *convexity* (pairwise) bound, Eq. (15.34), $I(Z; J) \leq M^{-2} \sum_{j,k} D(\mathcal{N}(0, \Sigma^j) \parallel \mathcal{N}(0, \Sigma^k))$, evaluated by Sherman–Morrison: with $\Sigma^j = I + \nu\theta_j\theta_j^\top$ ($\|\theta_j\| = 1$, so $\det \Sigma^j = 1 + \nu$ and the log-determinants cancel), $D = \frac{1}{2}\nu^2(1 - \langle\theta_j, \theta_k\rangle^2)/(1 + \nu)$, and the tilt construction gives $1 - \langle\theta_j, \theta_k\rangle^2 \leq 4\delta^2$ *per pair*. The crucial gain: the pairwise sum $\sum_{j,k}(1 - \langle\theta_j, \theta_k\rangle^2) \leq 4\delta^2 M^2$ needs *no* $\bar{v} = 0$ /symmetry and *no* block-diagonal Gaussian bridge—it eliminates both the isotropy averaging and the $n \cdot d$ -dimensional reduction:

```
refine le_trans (mutualInformation_le_avg_pairwise_kl Qc) ?_ -- Eq. (15.34), not Lemma 15.17
rw [klDiv_multivariateGaussian_zero _ _ (spiked_posDef ua h.le) (spiked_posDef ub h.le), ...]
```

The price is a worse constant (2048^{-1} , §??); the rate $\min\{(1+\nu)/\nu^2 \cdot d/n, 1\}$ is the book’s. **Cause:** (iv).

6 The removed result: Example 15.11 (optimal $n^{-4/9}$)

Example 15.11 is the only result we *removed*. Its honest proof needs the convex-hull (mixture) TV bound between the uniform U^n and the sign-vector mixture $Q = 2^{-m} \sum_{\alpha \in \{\pm 1\}^m} P_\alpha^n$. Wainwright himself *does not prove* this—he cites it: “Birgé and Massart (1995) prove the upper bound (15.28) on the squared Hellinger distance; see theorem 1 in their paper.” Formalizing (15.28) (or our cleaner χ^2 -second-moment substitute, $\chi^2(Q \parallel U^n) + 1 = \mathbb{E}_\epsilon[(1 + \sum_j \epsilon_j b_j)^n] \leq e^{(n^2/2) \sum_j b_j^2}$) is a from-scratch χ^2 -over-product-measure development with no Mathlib precedent; it resisted four cluster sessions.

What forced the decision: the existential statement $\exists c > 0$, $\text{ofReal}(c \cdot n^{-4/9}) \leq \text{minimaxRiskDist id } \theta P$ has the constant c quantified *after* n , so it is satisfiable *vacuously*—a fixed two-point family $\{U, 1 + \frac{1}{2} \sin(2\pi x)\}$ gives a fixed $(\delta/2)(1 - \text{TV}) > 0$, after which “a small enough $c > 0$ is always available.” That 0-**sorry** “proof” establishes only “positive minimax risk for each n ,” *not* the $n^{-4/9}$ scaling. Rather than ship a vacuous rate or leave a research-grade **sorry**, the example file was deleted (and its umbrella import removed). The suboptimal two-point rate $n^{-1/2}$ (Example 15.8) remains, faithfully.

Methodological note (applies to all existential-rate statements). The pattern “ $\forall n, \exists c > 0, c \cdot n^{-r} \leq \text{risk}$ ” is gameable. The other examples avoid the trap because their constructions deliver a *uniform* c (an explicit numeric constant, e.g. Ex 15.4’s $v/24n$) or because the method (functional modulus, with $H^2 \leq 1/(4n)$) forces an n -scaled construction; only the convex-hull crux of Ex 15.11 admitted the fixed-family shortcut.

7 Conclusion

Of eight revised/removed results, six are not attributable to any error in Wainwright: four are asymptotic-vs-finite edge guards or implicit positivity (Examples 15.5, 5.12, 15.14, 15.19; Lemma

15.17), and two are tractability proof-substitutions plus loose constants (Lemma 15.17 and Example 15.19, avoiding differential entropy and sphere isotropy). Exactly two trace to genuine informal looseness in the book—the squared-vs-absolute loss convention of Examples 15.8/15.11, and the implicit richness of “the class.” One result, Example 15.11, relies on a bound the book itself only cites (Birgé–Massart, 1995) and was removed rather than faked. In all cases the book’s *mathematics* is correct; the revisions are about making an informal, asymptotic argument into a precise, finite, machine-checked one with the bricks Mathlib currently provides.